

BMAT202L

Probability & Statistics

Complete FAT Preparation Guide

Based on 13 Previous Year Question Papers

2021-22 | 2022-23 | 2023-24 | 2024-25

All Slots: B1, B2, D1, D2, G1

30+
Question Types

VIT Vellore • BMAT202L FAT • April 2025

HOW TO USE THIS GUIDE

Each topic has multiple TYPES.	Each TYPE = one distinct question pattern that has appeared in PYQs.
Every TYPE includes:	Recognition cue → Formulas → Step-by-step method → Fully solved PYQ examples → Key differences from similar types.
Colour coding:	Green box = how to recognise Blue box = formulas Teal box = solved examples Red box = warnings Orange box = tips.
Highest-priority questions:	Marked with a star. These appeared 4+ times across all papers — know them cold.
End of guide:	A compact Variation Summary Table maps every type to its key trick for last-minute revision.

TOPIC 1

Descriptive Statistics

Mean · Median · Mode · Standard Deviation · Quartile Deviation · CV · Skewness

TYPE 1 — Mean, Median & Mode from Grouped Frequency Distribution

HOW TO RECOGNISE: Table with class intervals + frequency. Asked for Mean, Median and/or Mode.

Key Formulas

Mean	$\bar{x} = \Sigma(f \times m) / N$ where m = class midpoint, $N = \Sigma f$
Median	$\text{Median} = L + [(N/2 - CF) / f] \times h$
Mode	$\text{Mode} = L + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$
	L = lower boundary of median/modal class
	CF = cumulative freq before that class, f = class freq, h = class width
	f_0 = pre-modal freq, f_1 = modal freq, f_2 = post-modal freq

Step-by-Step Method

- 1 Compute midpoints: $m = (\text{lower} + \text{upper}) / 2$ for each class.
- 2 Build columns: f , m , $f \times m$, $f \times m^2$. Sum all: get $\Sigma f = N$, Σfm , Σfm^2 .
- 3 $\text{Mean} = \Sigma fm / N$.
- 4 Build cumulative frequency column. Find $N/2$. Median class = first class where $CF \geq N/2$. Apply formula.
- 5 Modal class = class with highest frequency. Apply Mode formula.

Solved PYQ Examples

SOLVED EXAMPLE — Apr 2023 G1 — Fracture Strength of Ceramic Bars

Classes: 81-83, 83-85, 85-87, 87-89, 89-91, 91-93, 93-95, 95-97, 97-99

Freq: 6, 7, 17, 30, 43, 28, 22, 13, 3 N = 169

Step 1: Midpoints = 82, 84, 86, 88, 90, 92, 94, 96, 98

$\Sigma fm = 6 \times 82 + 7 \times 84 + 17 \times 86 + 30 \times 88 + 43 \times 90 + 28 \times 92 + 22 \times 94 + 13 \times 96 + 3 \times 98$

$\Rightarrow = 492 + 588 + 1462 + 2640 + 3870 + 2576 + 2068 + 1248 + 294 = 15238$

$\Rightarrow \text{Mean} = 15238 / 169 = 90.16 \text{ MPa}$

Step 2: Cumulative freq: 6, 13, 30, 60, 103, 131, 153, 166, 169

$N/2 = 84.5 \rightarrow \text{Median class} = 89-91$ (CF before = 60, $f = 43$, $h = 2$, $L = 89$)

$\Rightarrow \text{Median} = 89 + [(84.5 - 60) / 43] \times 2 = 89 + 1.14 = 90.14 \text{ MPa}$

Step 3: Modal class = 89-91 ($f=43$ highest); $f_0=30$, $f_1=43$, $f_2=28$

$\Rightarrow \text{Mode} = 89 + [(43-30) / (86-30-28)] \times 2 = 89 + [13/28] \times 2 = 89.93 \text{ MPa}$

SOLVED EXAMPLE — Apr 2025 D1 — Homes Visited by Salesman

Class: 0-9, 10-19, 20-29, 30-39, 40-49

Freq: 3, 8, 24, 60, 21 N = 116

Midpoints: 4.5, 14.5, 24.5, 34.5, 44.5

$\Rightarrow \Sigma fm = 13.5 + 116 + 588 + 2070 + 934.5 = 3722$

$\Rightarrow \text{Mean} = 3722 / 116 = 32.09$

Cumulative freq: 3, 11, 35, 95, 116

$N/2 = 58 \rightarrow \text{Median class} = 30-39$ (CF=35, $f=60$, $h=10$, $L=30$)

$\Rightarrow \text{Median} = 30 + [(58-35)/60] \times 10 = 30 + 3.83 = 33.83$

Modal class = 30-39; $f_0=24$, $f_1=60$, $f_2=21$

$\Rightarrow \text{Mode} = 30 + [(60-24)/(120-24-21)] \times 10 = 30 + [36/75] \times 10 = 34.8$

SOLVED EXAMPLE — Apr 2025 G1 — 8-Class Distribution

Class: 0-10, 10-20, 20-30, 30-40, 40-50, 50-60, 60-70, 70-80

Freq: 5, 8, 7, 12, 28, 20, 10, 10 N = 100

Midpoints: 5, 15, 25, 35, 45, 55, 65, 75

$\Rightarrow \Sigma fm = 25+120+175+420+1260+1100+650+750 = 4500$

$\Rightarrow \text{Mean} = 4500/100 = 45$

CF: 5,13,20,32,60,80,90,100 $\rightarrow N/2=50 \rightarrow \text{Median class} = 40-50$

$\Rightarrow \text{Median} = 40 + [(50-32)/28] \times 10 = 40 + 6.43 = 46.43$

Modal class = 40-50 ($f=28$); $f_0=12$, $f_1=28$, $f_2=20$

$\Rightarrow \text{Mode} = 40 + [(28-12)/(56-12-20)] \times 10 = 40 + [16/24] \times 10 = 46.67$

WARNING: Always use class BOUNDARIES for L (not limits). For 81-83: L=81. Do NOT confuse midpoint with L.

TYPE 2 — SD, Quartile Deviation, CV & Skewness from Grouped Data

HOW TO RECOGNISE: Wage/salary table. Multiple measures asked in one question: SD, QD, CV, Skewness.

Key Formulas

SD	$\sigma = \sqrt{[\sum f(m - x)^2 / N]} \text{ or } \sqrt{[\sum fm^2/N - (x)^2]}$
Q1 / Q3	$Q_i = L + [(iN/4 - CF) / f] \times h$
Quartile Dev.	$QD = (Q3 - Q1) / 2$
CV	$CV = (\sigma / x) \times 100\%$
Pearson Skewness	$Sk = (Mean - Mode) / \sigma \text{ or } 3(Mean - Median) / \sigma$
Bowley's Skewness	$Sk = (Q3 + Q1 - 2 \times Median) / (Q3 - Q1)$

SOLVED EXAMPLE — Apr 2023 G1 — Monthly Wages of 72 Workers

Classes: 12.5-17.5, 17.5-22.5, 22.5-27.5, 27.5-32.5, 32.5-37.5,

37.5-42.5, 42.5-47.5, 47.5-52.5, 52.5-57.5

Freq: 2, 22, 19, 14, 3, 4, 6, 1, 1 N=72

Mean calculation:

$$\Sigma fm = 2 \times 15 + 22 \times 20 + 19 \times 25 + 14 \times 30 + 3 \times 35 + 4 \times 40 + 6 \times 45 + 1 \times 50 + 1 \times 55$$

$$= 30 + 440 + 475 + 420 + 105 + 160 + 270 + 50 + 55 = 2005$$

$$\Rightarrow \bar{x} = 2005/72 = 27.85$$

SD calculation:

$$\Sigma fm^2 = 2 \times 225 + 22 \times 400 + 19 \times 625 + 14 \times 900 + 3 \times 1225 + 4 \times 1600 + 6 \times 2025 + 1 \times 2500 + 1 \times 3025 = 61475$$

$$\Rightarrow \sigma^2 = 61475/72 - (27.85)^2 = 854.51 - 776.02 = 78.49$$

$$\Rightarrow \sigma = 8.86$$

Quartiles: CF: 2, 24, 43, 57, 60, 64, 70, 71, 72

Q1: $N/4=18 \rightarrow$ class 17.5-22.5 (CF=2, f=22):

$$\Rightarrow Q1 = 17.5 + [(18-2)/22] \times 5 = 17.5 + 3.64 = 21.14$$

Q3: $3N/4=54 \rightarrow$ class 27.5-32.5 (CF=43, f=14):

$$\Rightarrow Q3 = 27.5 + [(54-43)/14] \times 5 = 27.5 + 3.93 = 31.43$$

$$\Rightarrow QD = (31.43 - 21.14)/2 = 5.15$$

$$CV = (8.86/27.85) \times 100 = 31.8\%$$

TYPE 3 — Find Missing Frequency Given Mean or Median

HOW TO RECOGNISE: Frequency table has one or more missing values (f3, f4...). Mean or Median is stated. Set up equations.

Strategy

- 1 Equation 1: Total frequency = N $\rightarrow$ sum all known + unknowns = N.
- 2 Equation 2: If Mean given: $\Sigma fm/N = \text{Mean} \rightarrow$ plug in unknowns and solve.
- 3 If Median given: Identify median class (need $N/2$ and CF), apply median formula, equate to given median.
- 4 If Mode given: Identify modal class (highest f), apply mode formula.
- 5 Solve the system of equations.

SOLVED EXAMPLE — B1 Apr 2025 — Missing Frequency, Mean = 28

Profit (Rs): 0-10, 10-20, 20-30, 30-40, 40-50, 50-60

Shops: 12, 18, 27, f, 17, 6

Equation 1 (total): $12+18+27+f+17+6 = 80+f$

Equation 2 (mean=28):

$\Sigma fm = 12 \times 5 + 18 \times 15 + 27 \times 25 + f \times 35 + 17 \times 45 + 6 \times 55 = 2100 + 35f$

$\Rightarrow (2100+35f)/(80+f) = 28$

$\Rightarrow 2100+35f = 2240+28f$

$\Rightarrow 7f = 140 \Rightarrow f = 20$

Now N=100. CF: 12,30,57,77,94,100

Median class: $N/2=50 \rightarrow 20-30$ (CF=30, f=27, h=10, L=20)

$\Rightarrow \text{Median} = 20 + [(50-30)/27] \times 10 = 20 + 7.41 = 27.41$

SOLVED EXAMPLE — Apr 2025 D2 — Median=33.50, Mode=34, Total=230

Wages: 0-10, 10-20, 20-30, 30-40, 40-50, 50-60, 60-70

Freq: 4, 16, f3, f4, f5, 6, 4

Eq 1 (total): $f3+f4+f5 = 200$

Mode=34 $\rightarrow$ modal class=30-40 $\rightarrow$ f4 is max.

Mode formula: $30 + [(f4-f3)/(2f4-f3-f5)] \times 10 = 34$

$\Rightarrow (f4-f3)/(2f4-f3-f5) = 0.4 \dots(2)$

Median=33.50 $\rightarrow$ median class=30-40

(check: $N/2=115$, CF before 30-40 = $20+f3$)

$\Rightarrow 33.5 = 30 + [(115-(20+f3))/f4] \times 10$

$\Rightarrow 0.35 \times f4 = 95-f3 \dots(3)$

Solve (1),(2),(3) simultaneously to find f3, f4, f5.

TYPE 4 — Coefficient of Variation to Compare Consistency

HOW TO RECOGNISE: Two teams/groups given. Compute Mean, SD, CV for each. Lower CV = more consistent.

TIP: $CV = (\sigma/\bar{x}) \times 100$. If $CV_A < CV_B$ then A is more consistent (less variable relative to its mean).

SOLVED EXAMPLE — May 2024 D2 — Football Teams A & B

Goals (x): 0, 1, 2, 3, 4, 5 with matches for Team A and Team B

For each team:

Step 1: Compute $\bar{x} = \Sigma(\text{goals} \times \text{matches}) / \Sigma \text{matches}$

Step 2: Compute $\sigma^2 = \Sigma(x^2 \times f)/N - (\bar{x})^2$, then $\sigma = \sqrt{\sigma^2}$

Step 3: $CV = (\sigma/\bar{x}) \times 100$

Team with LOWER CV is more consistent.

KEY DIFFERENCES

- TYPE 2 asks for SD + QD + CV + Skewness — many measures from one grouped table.
- TYPE 4 asks ONLY for CV, applied to TWO groups to compare — consistency is the point.
- Triggered differently: Type 2 by salary/wage tables; Type 4 by 'which team is more consistent?'

TOPIC 2

Random Variables — Discrete

PMF · CDF · Mean · Variance · Geometric Distribution · $Y = g(X)$

TYPE 1 — Find k, Mean, Variance, CDF from PMF Table

HOW TO RECOGNISE: A table of x values and p(x) with unknown k. Find k, then compute E(X), Var(X), CDF.

Key Formulas

Find k	$\sum p(x) = 1$
Mean	$E(X) = \sum x \cdot p(x)$
2nd Moment	$E(X^2) = \sum x^2 \cdot p(x)$
Variance	$\text{Var}(X) = E(X^2) - [E(X)]^2$
CDF	$F(x) = P(X \leq x) = \sum p(x) \text{ for all } x \leq x$

SOLVED EXAMPLE — B2 Apr 2025 — PMF with k (6 values)

x: -2 -1 0 1 2 3

p(x): 0.1 k 0.2 2k 0.3 k

Step 1 — Find k:

$$0.1 + k + 0.2 + 2k + 0.3 + k = 1$$

$$\Rightarrow 0.6 + 4k = 1 \Rightarrow k = 0.1$$

Updated PMF: p(-1)=0.1, p(1)=0.2, p(3)=0.1**Step 2 — Mean:**

$$E(X) = (-2)(0.1) + (-1)(0.1) + (0)(0.2) + (1)(0.2) + (2)(0.3) + (3)(0.1)$$

$$= -0.2 - 0.1 + 0 + 0.2 + 0.6 + 0.3 = 0.8$$

Step 3 — Variance:

$$E(X^2) = 4(0.1) + 1(0.1) + 0(0.2) + 1(0.2) + 4(0.3) + 9(0.1) = 0.4 + 0.1 + 0 + 0.2 + 1.2 + 0.9 = 2.8$$

$$\Rightarrow \text{Var}(X) = 2.8 - (0.8)^2 = 2.8 - 0.64 = 2.16$$

Step 4 — CDF:

$$F(-2)=0.1, F(-1)=0.2, F(0)=0.4, F(1)=0.6, F(2)=0.9, F(3)=1.0$$

SOLVED EXAMPLE — G1 Apr 2025 — PMF with 3K variant

x: -2 -1 0 1 2 3

p(x): 0.1 K 0.2 2K 0.3 3K

$$\Sigma p = 0.1 + K + 0.2 + 2K + 0.3 + 3K = 0.6 + 6K = 1 \Rightarrow K = 1/15 \approx 0.0667$$

$$E(X) = (-2)(0.1) + (-1)(1/15) + (0)(0.2) + (1)(2/15) + (2)(0.3) + (3)(3/15)$$

$$\Rightarrow = -0.2 - 0.0667 + 0 + 0.1333 + 0.6 + 0.6 = 1.067$$

$$P(X < 2) = F(1) = p(-2) + p(-1) + p(0) + p(1) = 0.1 + 1/15 + 0.2 + 2/15 = 0.5$$

$$P(-2 < X < 2) = F(1) - F(-2) = 0.5 - 0.1 = 0.4$$

SOLVED EXAMPLE — B1 Apr 2025 — Quadratic in k (7 values)

x: 0 1 2 3 4 5 6 7

p(x): 0 k 2k 2k 3k k² 2k² 7k²+k

$$\Sigma p = 9k + 10k^2 = 1 \Rightarrow 10k^2 + 9k - 1 = 0$$

$$\Rightarrow k = [-9 \pm \sqrt{(81+40)}] / 20 = [-9 \pm 11] / 20$$

$$\Rightarrow k = 0.1 \text{ (take positive root)}$$

$$P(X < 6) = p(1) + p(2) + p(3) + p(4) + p(5) = 0.1 + 0.2 + 0.2 + 0.3 + 0.01 = 0.81$$

$$P(X \geq 6) = 1 - 0.81 = 0.19$$

$$P(0 < X < 5) = 0.1 + 0.2 + 0.2 + 0.3 = 0.8$$

TYPE 2 — Geometric / Infinite Discrete Distribution

HOW TO RECOGNISE: $P(X=x)$ given as a formula: $(3/4)(1/4)^{(x-1)}$ or $1/2^x$. Find probabilities, mean, variance.

Key Formulas — Geometric Distribution $P(X=x) = p \cdot q^{(x-1)}$

Mean	$E(X) = 1/p$
Variance	$\text{Var}(X) = q / p^2$ where $q = 1 - p$
Memoryless	$P(X > a+b \mid X > a) = P(X > b)$ — KEY property!
$P(X > n)$	$P(X > n) = q^n$
Infinite sum	For $P(X=j)=1/2^j$: $E(X)=2$, $\text{Var}(X)=2$, $P(X \text{ even})=1/3$

SOLVED EXAMPLE — D1 Apr 2025 — Geometric, $p=3/4$

$$P(X=x) = (3/4)(1/4)^{x-1}, x = 1, 2, 3, \dots$$

$P(X>4)$:

$$P(X>4) = (1/4)^4 = 1/256 \text{ [since } P(X>n) = q^n]$$

$P(X>4 | X>2)$ — use MEMORYLESS property:

$$P(X>4|X>2) = P(X>2) = (1/4)^2 = 1/16$$

$$E(X) = 1/p = 1/(3/4) = 4/3$$

$$\text{Var}(X) = q/p^2 = (1/4)/(9/16) = 4/9$$

SOLVED EXAMPLE — B1 May 2024 — $P(X=j) = 1/2^j$

$$P(X=x) = 1/2^x, x = 1, 2, 3, \dots$$

$$P(X \text{ is even}) = P(X=2)+P(X=4)+\dots = 1/4+1/16+\dots$$

$$= (1/4) / (1-1/4) = (1/4)/(3/4) = 1/3$$

$$P(X \leq 5) = 1/2+1/4+1/8+1/16+1/32 = 31/32$$

$$P(X \text{ divisible by } 3) = P(X=3)+P(X=6)+\dots$$

$$= (1/8)/(1-1/64) = (1/8)/(63/64) = 8/63$$

$$E(X) = \sum x/2^x = 2 \text{ [standard result]}$$

$$\text{Var}(X) = E(X^2) - 4 = 6 - 4 = 2$$

TYPE 3 — Function of a Random Variable: $Y = g(X)$

HOW TO RECOGNISE: Given PMF of X , define $Y = X^2 + 2X$ or similar. Find PMF of Y and $E(Y)$.

Method

- 1 Compute Y value for each x : substitute into the formula.
- 2 Build PMF of Y : $P(Y=y) = \text{sum of } P(X=x) \text{ for all } x \text{ that give that } y \text{ value.}$
- 3 Find $E(Y) = \sum y \cdot P(Y=y)$ OR use $E(Y) = E(g(X)) = \sum g(x) \cdot P(X=x)$ directly.

SOLVED EXAMPLE — B2 Apr 2025 — $Y = X^2 + 2X$

x : 0 1 2 3

$p(x)$: 0.1 0.3 0.5 0.1

Compute $Y = x^2 + 2x$ for each x :

$x=0$: $Y=0$ | $x=1$: $Y=3$ | $x=2$: $Y=8$ | $x=3$: $Y=15$

PMF of Y :

$P(Y=0)=0.1$, $P(Y=3)=0.3$, $P(Y=8)=0.5$, $P(Y=15)=0.1$

$E(Y) = 0 \times 0.1 + 3 \times 0.3 + 8 \times 0.5 + 15 \times 0.1$

$\Rightarrow = 0 + 0.9 + 4 + 1.5 = 6.4$

Verify: $E(Y) = E(X^2 + 2X) = E(X^2) + 2E(X)$

$E(X) = 0 \times 0.1 + 1 \times 0.3 + 2 \times 0.5 + 3 \times 0.1 = 1.6$

$E(X^2) = 0 \times 0.1 + 1 \times 0.3 + 4 \times 0.5 + 9 \times 0.1 = 3.2$

$\Rightarrow E(Y) = 3.2 + 2(1.6) = 6.4 \checkmark$

TYPE 4 — Continuous Univariate PDF: Find k , CDF, Probabilities

HOW TO RECOGNISE: $f(x) = k \cdot x^n$ or $k(1-x)^2$ etc., over $[0,1]$. Integrate to find k , then compute everything.

Key Formulas

Normalise	$\int f(x) dx = 1$ over the support $\rightarrow$ find k
Probability	$P(a < X < b) = \int [a \text{ to } b] f(x) dx$
CDF	$F(x) = \int [0 \text{ to } x] f(t) dt$
Mean	$E(X) = \int x \cdot f(x) dx$
Variance	$\text{Var}(X) = E(X^2) - [E(X)]^2$ where $E(X^2) = \int x^2 \cdot f(x) dx$

SOLVED EXAMPLE — D1 May 2024 — $f(x) = 2x$, $0 < x < 1$

Verify: $\int [0 \text{ to } 1] 2x dx = [x^2]_0^1 = 1 \checkmark$

$P(X < 1/2) = \int [0 \text{ to } 1/2] 2x dx = [x^2]_0^{1/2} = 1/4$

$P(1/4 < X < 1/2) = (1/2)^2 - (1/4)^2 = 1/4 - 1/16 = 3/16$

$P(X > 3/4) = 1 - (3/4)^2 = 7/16$

Conditional: $P(X > 3/4 | X > 1/2) = P(X > 3/4) / P(X > 1/2) = (7/16) / (3/4) = 7/12$

SOLVED EXAMPLE — D2 Apr 2025 — $f(x) = 3(1-x)^2$, $0 \leq x \leq 1$ [Dealer's Profit]

$$E(X) = \int_0^1 x \cdot 3(1-x)^2 dx = 3 \int_0^1 (x - 2x^2 + x^3) dx$$

$$= 3 \left[\frac{1}{2} - \frac{2}{3} + \frac{1}{4} \right] = 3 \times \left(\frac{1}{12} \right) = \frac{1}{4}$$

$$E(X^2) = 3 \int_0^1 (x^2 - 2x^3 + x^4) dx = 3 \left[\frac{1}{3} - \frac{1}{2} + \frac{1}{5} \right] = 3 \left(\frac{1}{30} \right) = \frac{1}{10}$$

$$\Rightarrow \text{Var}(X) = \frac{1}{10} - \left(\frac{1}{4} \right)^2 = \frac{3}{80}; \text{SD} = \sqrt{\frac{3}{80}}$$

$$\text{CDF: } F(x) = \int_0^x 3(1-t)^2 dt = \left[-(1-t)^3 \right]_0^x = 1 - (1-x)^3$$

$$\text{Expected profit} = E(X) \times \text{Rs. } 3000 = \left(\frac{1}{4} \right) \times 3000 = \text{Rs. } 750$$

TOPIC 3

Joint Probability Distributions

Discrete Joint Tables · Continuous Joint PDFs · Marginals · Conditionals · Independence

NOTE: This topic appears in EVERY paper. The $f(x,y)=x^2+xy/3$ question appeared in 4+ papers — it is the #1 priority question.

TYPE 1 — Discrete Joint Distribution (Table Given)

HOW TO RECOGNISE: 2D table of $P(X=x, Y=y)$ values. Find marginals, conditional, covariance, correlation, independence.

Key Formulas

Marginal X	$P(X=x) = \sum_y P(X=x, Y=y)$ [sum across each ROW]
Marginal Y	$P(Y=y) = \sum_x P(X=x, Y=y)$ [sum down each COLUMN]
Conditional	$P(X=x Y=y) = P(X=x, Y=y) / P(Y=y)$
Independence	X,Y independent ■ $P(X=x, Y=y) = P(X=x) \cdot P(Y=y)$ for ALL (x,y)
Covariance	$\text{Cov}(X, Y) = E(XY) - E(X) \cdot E(Y)$ where $E(XY) = \sum \sum xy \cdot P(x, y)$
Correlation	$\rho = \text{Cov}(X, Y) / (\sigma_X \cdot \sigma_Y)$

SOLVED EXAMPLE — Apr 2023 G1 — Fair Coin (3 tosses): X=head/tail on 1st, Y=total heads

List all 8 outcomes and classify (X, Y):

HHH:(1,3) HHT:(1,2) HTH:(1,2) HTT:(1,1)

THH:(0,2) THT:(0,1) TTH:(0,1) TTT:(0,0) Each prob = 1/8

Joint Table:

Y=0 Y=1 Y=2 Y=3 | P(X=x)

X=0: 1/8 2/8 1/8 0 | 1/2

X=1: 0 1/8 2/8 1/8 | 1/2

P(Y=y): 1/8 3/8 3/8 1/8 | 1

$$E(X)=1/2, E(Y)=3/2$$

$$E(XY) = 1 \cdot 1 \cdot (1/8) + 1 \cdot 2 \cdot (2/8) + 1 \cdot 3 \cdot (1/8) = 1/8 + 4/8 + 3/8 = 1$$

$$\text{Cov}(X,Y) = 1 - (1/2)(3/2) = 1 - 3/4 = 1/4$$

$$\sigma_X = 1/2; \sigma_Y = \sqrt{3}/2$$

$$\rho = (1/4) / [(1/2)(\sqrt{3}/2)] = 1/\sqrt{3} \approx 0.577$$

X and Y are NOT independent (Cov $\neq$ 0).

SOLVED EXAMPLE — D2 Apr 2025 — Discrete Joint Table with Fractions

X=1 X=2 X=3

Y=1: 1/12 1/6 0

Y=2: 0 1/9 1/5

Y=3: 1/18 1/4 2/15

Marginals:

$$P(X=1) = 1/12 + 0 + 1/18 = 3/36 + 2/36 = 5/36$$

$$P(X=2) = 1/6 + 1/9 + 1/4 = 6/36 + 4/36 + 9/36 = 19/36$$

$$P(X=3) = 0 + 1/5 + 2/15 = 3/15 + 2/15 = 5/15 = 12/36$$

$$P(X \leq 2, Y=3) = P(1,3) + P(2,3) = 1/18 + 1/4 = 2/36 + 9/36 = 11/36$$

$$P(X+Y < 4) = P(1,1) + P(1,2) + P(2,1) = 1/12 + 0 + 1/6 = 3/12 = 1/4$$

$$P(X=1|Y=2) = P(1,2)/P(Y=2) = 0 / (1/9 + 1/5) = 0$$

TYPE 2 — Continuous Joint PDF over a Rectangle ★ HIGHEST PRIORITY

HOW TO RECOGNISE: $f(x,y) = x^2 + xy/3$ or $3/4 + xy$. Region is a RECTANGLE ($0 \leq x \leq a$, $0 \leq y \leq b$). Find marginals, probabilities.

Key Formulas

$$\text{Marginal } f_X \quad f_X(x) = \int f(x,y) dy \text{ [integrate over full y range]}$$

$$\text{Marginal } f_Y \quad f_Y(y) = \int f(x,y) dx \text{ [integrate over full x range]}$$

Conditional	$f(x y) = f(x,y) / f_Y(y)$
$P(X > a)$	$\int [a \text{ to max}] f_X(x) dx$
$P(Y < X)$	$\iint_{\{y < x\}} f(x,y) dy dx = \int_{\min}^1 \int_{\min}^x f(x,y) dy dx$
$P(Y < b X < a)$	$P(Y < b \text{ AND } X < a) / P(X < a)$ — double integral over intersection region
Independence	X, Y independent $\Rightarrow f(x,y) = f_X(x) \cdot f_Y(y)$

SOLVED EXAMPLE — ★ $f(x,y) = x^2 + xy/3$, $0 \leq x \leq 1$, $0 \leq y \leq 2$ [APPEARED IN 4+ PAPERS]

Marginals:

$$f_X(x) = \int_{[0 \text{ to } 2]} (x^2 + xy/3) dy = [x^2y + xy^2/6]_{y=0}^{y=2} = 2x^2 + 4x/6 = 2x^2 + 2x/3$$

$$f_Y(y) = \int_{[0 \text{ to } 1]} (x^2 + xy/3) dx = [x^3/3 + x^2y/6]_{x=0}^{x=1} = 1/3 + y/6$$

(i) $P(X > 1/2)$:

$$= \int_{[1/2 \text{ to } 1]} (2x^2 + 2x/3) dx = [2x^3/3 + x^2/3]_{x=1/2}^{x=1} = 2/3 + 1/3 - (2/24 + 1/12) = 1 - (1/12 + 1/12) = 1 - 1/6 = 5/6$$

$$= (2/3 + 1/3) - (2/24 + 1/12) = 1 - (1/12 + 1/12) = 1 - 1/6 = 5/6$$

(ii) $P(Y < X)$ [careful: y limit goes 0 to x, not 0 to 2]:

$$= \int_{\min}^1 \int_{\min}^x (x^2 + xy/3) dy dx = \int_{\min}^1 [x^2y + xy^2/6]_{y=0}^{y=x} dx$$

$$= \int_{\min}^1 (x^3 + x^3/6) dx = \int_{\min}^1 (7x^3/6) dx = [7x^4/24]_{x=0}^{x=1} = 7/24$$

(iii) $P(Y < 1/2 | X < 1/2)$:

$$P(X < 1/2) = \int_{\min}^{1/2} (2x^2 + 2x/3) dx = [2x^3/3 + x^2/3]_{x=0}^{x=1/2} = 1/12 + 1/12 = 1/6$$

$$P(Y < 1/2 \text{ AND } X < 1/2) = \int_{\min}^{1/2} \int_{\min}^{1/2} (x^2 + xy/3) dy dx$$

$$= \int_{\min}^{1/2} [x^2y + xy^2/6]_{y=0}^{y=1/2} dx = \int_{\min}^{1/2} (x^2/2 + x/24) dx$$

$$= [x^3/6 + x^2/48]_{x=0}^{x=1/2} = 1/48 + 1/192 = 5/192$$

$$\Rightarrow P(Y < 1/2 | X < 1/2) = (5/192) / (1/6) = 5/32$$

(iv) Means: $E(X) = \int_{\min}^1 x \cdot (2x^2 + 2x/3) dx = \int_{\min}^1 (2x^3 + 2x^2/3) dx = [x^4/2 + 2x^3/9]_{x=0}^{x=1} = 1/2 + 2/9 = 13/18$

$$E(Y) = \int_{\min}^2 y \cdot (1/3 + y/6) dy = [y^2/6 + y^3/18]_{y=0}^{y=2} = 4/6 + 8/18 = 2/3 + 4/9 = 10/9$$

SOLVED EXAMPLE — D1 Apr 2025 — $f(x,y) = 3/4 + xy$, $0 < x < 1$, $0 < y < 1$

Marginals:

$$f_X(x) = \int_0^1 (3/4 + xy) dy = 3/4 + x/2$$

$$f_Y(y) = \int_0^1 (3/4 + xy) dx = 3/4 + y/2$$

$$\text{Conditional: } f(x|y) = (3/4 + xy)/(3/4 + y/2)$$

$$P(Y > 1/2) = \int_{1/2}^1 (1/2)^1 (3/4 + y/2) dy = [3y/4 + y^2/4]_{1/2}^1 = 1 - 7/16 = 9/16$$

Independence check:

$$f_X(x) \cdot f_Y(y) = (3/4 + x/2)(3/4 + y/2) = 9/16 + 3y/8 + 3x/8 + xy/4$$

$$f(x,y) = 3/4 + xy$$

These are NOT equal $\Rightarrow$ X and Y are NOT independent.

TYPE 3 — Continuous Joint PDF over a Triangle ($x < y$ region)

HOW TO RECOGNISE: $f(x,y)$ defined with region $0 < x < y < 1$. The inner integral limits **DEPEND ON** x or y. Must sketch!

WARNING: CRITICAL: When region is $0 < x < y < 1$, for marginal $f_X(x)$: y goes from x to 1 (NOT 0 to 1). For $f_Y(y)$: x goes from 0 to y. Always sketch the triangular region first!

$$\text{Marginal } f_X(x) \quad f_X(x) = \int_{[x \text{ to } 1]} f(x,y) dy \text{ [y ranges from x up to 1]}$$

$$\text{Marginal } f_Y(y) \quad f_Y(y) = \int_{[0 \text{ to } y]} f(x,y) dx \text{ [x ranges from 0 up to y]}$$

$$P(Y > b | X = a) \quad \int_{[b \text{ to } 1]} f(y|x=a) dy \text{ where } f(y|x) = f(x,y)/f_X(x)$$

SOLVED EXAMPLE — G1 Apr 2025 — $f(x,y) = 10xy^2$, $0 < x < y < 1$
Marginal $f_X(x)$ [y: x to 1]:

$$f_X(x) = \int_{[x \text{ to } 1]} 10xy^2 dy = 10x[y^3/3]_x^1 = (10x/3)(1-x^3)$$

Marginal $f_Y(y)$ [x: 0 to y]:

$$f_Y(y) = \int_{[0 \text{ to } y]} 10xy^2 dx = 10y^2[x^2/2]_0^y = 5y^3$$

Conditional $f(y|x)$:

$$f(y|x) = 10xy^2 / [(10x/3)(1-x^3)] = 3y^2 / (1-x^3)$$

 $P(Y > 1/2 | X = 0.25)$:

$$= \int_{1/2}^1 3y^2 / (1-0.25^3) dy = [y^3/(1-1/64)]_{1/2}^1$$

$$= (1-1/8)/(63/64) = (7/8) \times (64/63) = 448/504 = 8/9$$

SOLVED EXAMPLE — B1 Apr 2025 — $f(x,y) = (1/8)(6-x-y)$, $0 < x < 2$, $2 < y < 4$ [Rectangular but complex limits]

P(X<1, Y<3):

$$= \int_{-1}^1 \int_{-3}^3 (1/8)(6-x-y) \, dy \, dx$$

$$\text{Inner: } (1/8)[6y - xy - y^2/2]_{-3}^3 = (1/8)[(18 - 3x - 9/2) - (12 - 2x - 2)]$$

$$= (1/8)[13.5 - 3x - (10 - 2x)] = (1/8)(3.5 - x)$$

$$\text{Outer: } (1/8) \int_{-1}^1 (3.5 - x) \, dx = (1/8)[3.5x - x^2/2]_{-1}^1 = (1/8)(3) = 3/8$$

P(X+Y < 3) [note: since y>2, X+Y<3 means X<1 AND Y<3-X]:

Region: 0<x<1, 2≤y so x<1

$$= \int_{-1}^1 \int_{2-x}^3 (1/8)(6-x-y) \, dy \, dx$$

KEY DIFFERENCES

- TYPE 2 (rectangle): both limits are CONSTANTS. e.g. $0 \leq x \leq 1$, $0 \leq y \leq 2$.
- TYPE 3 (triangle): at least one limit DEPENDS ON the other variable. e.g. $0 < x < y < 1$.
- Recognition: if you see $0 < x < y < 1$ or $x < y$ in the region description → Type 3, sketch first.
- P(Y<X) in a rectangular region is also a triangular integral (y goes 0 to x).

TOPIC 4

Correlation & Regression

Pearson's r · Rank Correlation · Regression Lines · Destroyed Lab Record · Multiple Regression

TYPE 1 — Karl Pearson's Coefficient of Correlation

HOW TO RECOGNISE: Two variables x , y in a table. Find r . May also ask to estimate y for a given x value.

Key Formula

$$r = \frac{[n \cdot \Sigma xy - \Sigma x \cdot \Sigma y]}{\sqrt{[(n \cdot \Sigma x^2 - (\Sigma x)^2) \cdot (n \cdot \Sigma y^2 - (\Sigma y)^2)]}}$$

Table needed **Make columns: x , y , x^2 , y^2 , xy . Sum all 5 columns. Plug into formula.**

Regression y on x **$\bar{y} - b_{yx} \cdot (\bar{x} - x)$ $b_{yx} = \frac{[n \Sigma xy - \Sigma x \Sigma y]}{[n \Sigma x^2 - (\Sigma x)^2]}$**

SOLVED EXAMPLE — Apr 2023 G1 — Infant Weight vs Chest Size ($n=9$) x (weight): 2.75, 2.15, 4.41, 5.52, 3.21, 4.32, 2.31, 4.30, 3.71 y (chest): 29.5, 26.3, 32.2, 36.5, 27.2, 27.7, 28.3, 30.3, 28.7**Compute sums ($n=9$):** $\Sigma x \approx 32.68$, $\Sigma y \approx 266.7$, $\Sigma x^2 \approx 130.36$, $\Sigma y^2 \approx 7919.9$, $\Sigma xy \approx 980.6$ **Numerator:** $9(980.6) - (32.68)(266.7) = 8825.4 - 8715.96 = 109.44$ **Denom factor 1:** $9(130.36) - (32.68)^2 = 1173.24 - 1067.98 = 105.26$ **Denom factor 2:** $9(7919.9) - (266.7)^2 = 71279.1 - 71128.89 = 150.21$ **$\Rightarrow r = 109.44 / \sqrt{(105.26 \times 150.21)} = 109.44 / 125.7 \approx 0.87$ [strong positive]****Regression line y on x :** $b_{yx} = 109.44 / 105.26 \approx 1.04$ $\bar{y} = 29.63$, $\bar{x} = 3.63$ **$\Rightarrow y - 29.63 = 1.04(x - 3.63) \rightarrow y = 1.04x + 25.85$** At $x=5$: $y = 1.04(5) + 25.85 = 31.05$ cm

TYPE 2 — Spearman's Rank Correlation

HOW TO RECOGNISE: Data in ranks OR raw data. Find ρ . With ties, use correction factor CF.

Key Formulas

No ties	$\rho = 1 - 6\sum D^2 / [n(n^2-1)]$ $D = R_x - R_y$ for each pair
With ties	$\rho = 1 - 6[\sum D^2 + \sum CF_x + \sum CF_y] / [n(n^2-1)]$
CF per tie	$CF = m(m-1) / 12$ $m = \text{number of tied observations}$
Assign ranks	Sort descending. Tied values get the AVERAGE of their rank positions.

SOLVED EXAMPLE — B1 2021-22 — x: 68,64,75,50,64,80,75,40,55,64 (n=10, with ties)

Rank x (sorted descending):

80→1, 75→2.5(tied), 75→2.5, 68→4, 64→6(tied×3), 64→6, 64→6, 55→8, 50→9, 40→10

Tie corrections for x:

64 appears 3 times → $m=3$: $CF = 3(9-1)/12 = 2$

75 appears 2 times → $m=2$: $CF = 2(4-1)/12 = 0.5$

$\sum CF_x = 2 + 0.5 = 2.5$

Similarly rank y, compute $D = R_x - R_y$ for each pair, $\sum D^2$

$$\rho = 1 - 6[\sum D^2 + 2.5 + \sum CF_y] / [10(99)]$$

SOLVED EXAMPLE — D2 Apr 2025 — x: 92,89,87,86,86,77,71,63,53,50 (n=10)

Rank x: 92→1, 89→2, 87→3, 86→4.5(tied), 86→4.5, 77→6, 71→7, 63→8, 53→9, 50→10

y: 86,83,91,77,68,85,52,82,37,57

Rank y: 91→1, 86→2, 85→3, 83→4, 82→5, 77→6, 68→7, 57→8, 52→9, 37→10

D values = $R_x - R_y$ for each pair:

D: 1-2=-1, 2-4=-2, 3-1=2, 4.5-6=-1.5, 4.5-7=-2.5, 6-3=3, 7-9=-2, 8-5=3, 9-10=-1, 10-8=2

$$\sum D^2 = 1+4+4+2.25+6.25+9+4+9+1+4 = 44.5$$

CF for x (86 tied twice) = $2(3)/12 = 0.5$

$$\rho = 1 - 6(44.5+0.5) / [10 \times 99] = 1 - 6(45)/990 = 1 - 270/990 = 0.727$$

TYPE 3 — Two Lines of Regression from Raw Data

HOW TO RECOGNISE: Data table. Find BOTH regression lines (y on x AND x on y). Use to estimate values.

Key Formulas

b_{yx}	$b_{yx} = [n\sum xy - \sum x \sum y] / [n\sum x^2 - (\sum x)^2]$ [regression y on x]
b_{xy}	$b_{xy} = [n\sum xy - \sum x \sum y] / [n\sum y^2 - (\sum y)^2]$ [regression x on y]
Line y on x	$y - \bar{y} = b_{yx} (x - \bar{x})$

Line x on y

$$x - x_{\bar{}} = b_{xy} (y - \bar{y})$$

Check

$$b_{yx} \times b_{xy} = r^2 \text{ (both regression coefficients have same sign as } r \text{)}$$

SOLVED EXAMPLE — B2 Apr 2025 — Price (X) vs Units Sold (Y) (n=7)

X: 34, 36, 32, 35, 30, 38, 40

Y: 3, 7, 6, 5, 9, 2, 8

Compute sums:

$$\Sigma x = 245, \Sigma y = 40, \Sigma x^2 = 8649, \Sigma y^2 = 268, \Sigma xy = 1345, x_{\bar{}} = 35, \bar{y} = 40/7 \approx 5.71$$

$$b_{yx} = [7(1345) - 245(40)] / [7(8649) - 245^2] = [9415 - 9800] / [60543 - 60025]$$

$$\Rightarrow = -385/518 = -0.743$$

$$b_{xy} = [7(1345) - 245(40)] / [7(268) - 40^2] = -385/276 = -1.395$$

$$\text{Line y on x: } y - 5.71 = -0.743(x - 35) \rightarrow y = -0.743x + 31.72$$

$$\text{Line x on y: } x - 35 = -1.395(y - 5.71) \rightarrow x = -1.395y + 42.97$$

Estimates:

$$\text{When } X=37: y = -0.743(37) + 31.72 = 4.23 \text{ units}$$

$$\text{When } Y=4: x = -1.395(4) + 42.97 = 37.39$$

TYPE 4 — Regression from Destroyed Lab Record ★ APPEARED IN 3+ PAPERS
HOW TO RECOGNISE: Given: $\text{Var}(X)=9$, two regression equations as expressions. Find means, r , σ_Y . EXACT SAME problem in G1, D1 2025 and B1 May 2024.

NOTE: The equations: $8X - 10Y + 66 = 0$ and $40X - 18Y - 214 = 0$ (or equivalent). This appeared in multiple slots.

- 1 Both regression lines pass through $(x_{\bar{}}, \bar{y})$. Substitute $x_{\bar{}}$, $\bar{y}$ and solve simultaneously.
- 2 Rewrite both equations as 'Y = ...' form. The one with smaller slope is y on x (b_{yx}); the other is x on y.
- 3 $r^2 = b_{yx} \times b_{xy}$, so $r = \sqrt{b_{yx} \times b_{xy}}$. Sign of r = sign of regression coefficients.
- 4 Use $b_{yx} = r \cdot \sigma_Y / \sigma_X$ and $\sigma_X = \sqrt{\text{Var}(X)} = 3$ to find σ_Y .

SOLVED EXAMPLE — ★ G1/D1 Apr 2025, B1 May 2024 — Destroyed Lab Record

Given: $\text{Var}(X)=9 \rightarrow \sigma_X=3$

Equations: $8X-10Y+66=0 \dots(1)$

$40X-18Y-214=0 \dots(2)$

Step 1 — Find means:

$8x - 10y + 66 = 0 \dots(1')$

$40x - 18y - 214 = 0 \dots(2')$

Multiply (1') by 5: $40x - 50y + 330 = 0$

Subtract (2'): $-50y + 18y = -214 - 330 \rightarrow -32y = -544 \rightarrow y = 17$

Sub in (1'): $8x = 170 - 66 = 104 \rightarrow x = 13$

Step 2 — Identify lines:

Eq (1) as Y: $Y = 0.8X + 6.6 \rightarrow b_{yx} = 0.8$

Eq (2) as X: $X = 0.45Y + 5.35 \rightarrow b_{xy} = 0.45$

Step 3 — Correlation:

$r^2 = b_{yx} \times b_{xy} = 0.8 \times 0.45 = 0.36 \rightarrow r = 0.6$ (positive)

Step 4 — σ_Y :

$b_{yx} = r \cdot \sigma_Y / \sigma_X \rightarrow 0.8 = 0.6 \cdot \sigma_Y / 3 \rightarrow \sigma_Y = 4$

TYPE 5 — Multiple Linear Regression (y on x1 and x2)

HOW TO RECOGNISE: Two independent variables x1, x2 and dependent y. Set up 3 normal equations and solve.

Normal Equations

Model	$y = b_0 + b_1x_1 + b_2x_2$
Eq 1	$\sum y = n \cdot b_0 + b_1 \cdot \sum x_1 + b_2 \cdot \sum x_2$
Eq 2	$\sum x_1 y = b_0 \cdot \sum x_1 + b_1 \cdot \sum x_1^2 + b_2 \cdot \sum x_1 x_2$
Eq 3	$\sum x_2 y = b_0 \cdot \sum x_2 + b_1 \cdot \sum x_1 x_2 + b_2 \cdot \sum x_2^2$
Predict	Plug x_1, x_2 values into fitted model

SOLVED EXAMPLE — Apr 2023 G1 — Wear (y) on Viscosity (x1) and Load (x2)

y: 193,172,113,230,91,125 x1: 16,22,33,15.5,43,40 x2: 851,1058,1357,816,1201,1115

Compute: $\Sigma y=924$, $\Sigma x1=169.5$, $\Sigma x2=6398$

Compute: $\Sigma x1y$, $\Sigma x2y$, $\Sigma x1^2$, $\Sigma x2^2$, $\Sigma x1x2$ ($5 \times 6 = 30$ multiplications)

Set up 3 equations, solve for $b0$, $b1$, $b2$

Predict at $x1=20$, $x2=1200$: $\blacksquare = b0+20b1+1200b2$

TOPIC 5

Standard Probability Distributions

Binomial · Poisson · Normal · Weibull · Exponential

TYPE 1 — Binomial Distribution $B(n, p)$

HOW TO RECOGNISE: n independent trials, each with success probability p . Discrete. Find $P(X=r)$ or $P(X \leq r)$ etc.

Key Formulas

<i>PMF</i>	$P(X=r) = C(n,r) \cdot p^r \cdot q^{(n-r)} \quad q = 1-p$
<i>Mean</i>	$E(X) = np$
<i>Variance</i>	$Var(X) = npq$
<i>At least 1</i>	$P(X \geq 1) = 1 - P(X=0) = 1 - q^n$

SOLVED EXAMPLE — B2 Apr 2025 — 800 Families with 5 Children, $p=0.5$

$n=5$, $p=q=0.5$, 800 families. $E(X) = np = 2.5$

(a) 3 boys: $P(X=3) = C(5,3)(0.5)^5 = 10/32 = 5/16$

Expected families = $800 \times 5/16 = 250$

(b) 5 girls [$X=0$ boys]: $P(X=0) = (0.5)^5 = 1/32$

Expected = $800/32 = 25$

(c) 2 or 3 boys: $P(X=2)+P(X=3) = [C(5,2)+C(5,3)](0.5)^5 = 20/32$

Expected = $800 \times 20/32 = 500$

(d) At least 1 boy: $P(X \geq 1) = 1 - (0.5)^5 = 31/32$

Expected = $800 \times 31/32 = 775$

SOLVED EXAMPLE — Apr 2023 G1 — Circuit Boards, $n=25$, $p=0.05$

$X \sim B(25, 0.05), q=0.95$

$$P(X=0) = (0.95)^{25} = 0.2774$$

$$P(X=1) = 25(0.05)(0.95)^{24} = 0.3650$$

$$P(X=2) = C(25,2)(0.05)^2(0.95)^{23} = 0.2305$$

$$P(X=3) = C(25,3)(0.05)^3(0.95)^{22} = 0.0930$$

$$P(X=4) = C(25,4)(0.05)^4(0.95)^{21} = 0.0269$$

$$P(X \leq 4) = 0.2774 + 0.3650 + 0.2305 + 0.0930 + 0.0269 = 0.9928$$

$$P(X > 5) = 1 - P(X \leq 5)$$

$$E(X) = 25 \times 0.05 = 1.25, SD = \sqrt{1.1875} = 1.09$$

TYPE 2 — Poisson Distribution $Poisson(\lambda)$

HOW TO RECOGNISE: Rare events with known mean rate λ . Find $P(X=k)$. If λ not given directly, estimate from data first.

Key Formulas

PMF	$P(X=k) = e^{(-\lambda)} \cdot \lambda^k / k!$
Mean = Var	$E(X) = Var(X) = \lambda$
Key values	$e^{(-1)} \approx 0.3679, e^{(-2)} \approx 0.1353, e^{(-3)} \approx 0.0498, e^{(-5)} \approx 0.0067$
Goodness of fit	Estimate $\lambda = \Sigma(k \cdot f_k) / N$, compute expected $E_k = N \cdot e^{(-\lambda)} \cdot \lambda^k / k!$, run χ^2 test

SOLVED EXAMPLE — D1 Apr 2025 — Tool Failures, $\lambda=1.25/\text{day}$, 4 days, 5 spares

Total λ over 4 days = 5. $X \sim Poisson(5)$

P(3 replacements over 4 days):

$$P(X=3) = e^{(-5)} \times 5^3 / 6 = 0.006738 \times 125 / 6 = 0.1404$$

Expected replacements made = $E[\min(X,5)]$:

$$= \sum_{k=0}^4 k \cdot P(X=k) + 5 \cdot P(X \geq 5)$$

$$= 0 + 1 \cdot P(1) + 2 \cdot P(2) + 3 \cdot P(3) + 4 \cdot P(4) + 5 \cdot [1 - \sum_{k=0}^4 P(k)]$$

$$P(0)=0.00674, P(1)=0.03369, P(2)=0.08422, P(3)=0.14037, P(4)=0.17547$$

$$= 0 + 0.03369 + 0.16844 + 0.42111 + 0.70188 + 5 \times (1 - 0.44149)$$

$$= 1.32512 + 5 \times 0.55851 = 1.325 + 2.793 = 4.118$$

TYPE 3 — Normal Distribution $N(\mu, \sigma^2)$

HOW TO RECOGNISE: μ and σ given. Standardise: $Z=(X-\mu)/\sigma$. Use standard normal table. Multiple sub-types below.

Key Formula & Z-Table Facts

Standardise	$Z = (X - \mu) / \sigma$
Common areas	$P(0 < Z < 1) = 0.3413$, $P(0 < Z < 2) = 0.4772$, $P(0 < Z < 1.96) = 0.4750$
	$P(0 < Z < 1.645) = 0.45$, $P(0 < Z < 2.326) = 0.49$, $P(0 < Z < 2.576) = 0.495$
Symmetry	$P(Z < -a) = P(Z > a) = 0.5 - P(0 < Z < a)$

SOLVED EXAMPLE — D2 Apr 2025 — Workers' Wages: $\mu=71$, $\sigma=6$, $n=900$

(i) Between Rs.65 and Rs.70:

$$Z_{65} = (65 - 71) / 6 = -1.0; Z_{70} = (70 - 71) / 6 = -0.167$$

$$P(65 < X < 70) = P(0 < Z < 1.0) - P(0 < Z < 0.167) = 0.3413 - 0.0662 = 0.2751$$

$$\text{Workers} = 900 \times 0.2751 \approx 248$$

(ii) More than Rs.74:

$$Z = (74 - 71) / 6 = 0.5$$

$$P(X > 74) = 0.5 - P(0 < Z < 0.5) = 0.5 - 0.1915 = 0.3085$$

$$\text{Workers} = 900 \times 0.3085 \approx 278$$

(iii) Less than Rs.60:

$$Z = (60 - 71) / 6 = -1.833$$

$$P(X < 60) = 0.5 - P(0 < Z < 1.833) = 0.5 - 0.4665 = 0.0335$$

$$\text{Workers} = 900 \times 0.0335 \approx 30$$

SOLVED EXAMPLE — Apr 2023 G1 — Birth Weights: $\mu=3432\text{g}$, $\sigma=482\text{g}$

(i) $P(X > 4000)$:

$$Z = (4000 - 3432) / 482 = 1.18$$

$$P(X > 4000) = 0.5 - 0.3810 = 0.1190$$

$P(3000 < X < 4000)$:

$$Z_{\blacksquare} = (3000 - 3432) / 482 = -0.896; Z_{\blacksquare} = 1.18$$

$$= P(0 < Z < 0.896) + P(0 < Z < 1.18) = 0.3150 + 0.3810 = 0.696$$

(ii) $P(X < 2000 \text{ or } X > 5000)$:

$$Z_{\blacksquare} = (2000 - 3432) / 482 = -2.97; Z_{\blacksquare} = (5000 - 3432) / 482 = 3.25$$

$$P(X < 2000) = 0.5 - 0.4985 = 0.0015$$

$$P(X > 5000) = 0.5 - 0.4994 = 0.0006$$

$$\text{Total} = 0.0021$$

SOLVED EXAMPLE — May 2024 D2 — Find μ and σ from Two Percentile Conditions

Given: 5% of men under 60 inches, 40% between 60 and 65 inches

$$P(X < 60) = 0.05 \rightarrow (60 - \mu) / \sigma = -1.645 \rightarrow \mu - 1.645\sigma = 60 \dots (1)$$

$$P(X < 65) = 0.45 \rightarrow (65 - \mu) / \sigma = -0.125 \rightarrow \mu - 0.125\sigma = 65 \dots (2)$$

Subtract (1) from (2):

$$1.52\sigma = 5 \rightarrow \sigma = 3.29$$

$$\Rightarrow \mu = 65 + 0.125(3.29) = 65.41 \text{ inches}$$

TOPIC 6

Hypothesis Testing

Z-test · t-test · F-test · Chi-Square · One-Way ANOVA · Two-Way ANOVA

Test	Critical value (5%)	Critical value (1%)	df
Z (two-tail)	$ Z > 1.96$	$ Z > 2.576$	—
Z (one-tail)	$ Z > 1.645$	$ Z > 2.326$	—
t-test	t-table at $\alpha=0.05$	t-table at $\alpha=0.01$	$n-1$ or n_1+n_2-2
F-test	F-table at $\alpha=0.05$	—	(n_1-1, n_2-1)
χ^2 -test	χ^2 -table at $\alpha=0.05$	—	$(r-1)(c-1)$

TYPE 1 — Z-test: Single Sample Mean (Large n)

HOW TO RECOGNISE: Sample size $n > 30$. Population mean μ claimed. Test whether sample supports the claim.*Z statistic*

$$Z = (x - \mu) / (\sigma/\sqrt{n}) \text{ or } (x - \mu) / (S/\sqrt{n}) \text{ if } \sigma \text{ unknown}$$

SOLVED EXAMPLE — B2 Apr 2025 — Mean Height, $n=100$, $x=160$, $\mu=165$, $\sigma=10$ $H_0: \mu=165$ $H_1: \mu \neq 165$ (two-tailed, 5% LOS)

$$Z = (160-165)/(10/\sqrt{100}) = -5/1 = -5$$

$$|Z|=5 > 1.96 \rightarrow \text{REJECT } H_0$$

Conclusion: The claim that mean height is 165 cm is not supported.

TYPE 2 — Z-test: Difference of Two Large Sample Means

HOW TO RECOGNISE: Two independent samples with large n_1, n_2 . Test if $\mu_1 = \mu_2$.*Z*

$$Z = (x_1 - x_2) / \sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$$

SOLVED EXAMPLE — Apr 2023 G1 — Electric Bulbs: Brand A ($n=1000$) vs Brand B ($n=75$) $x_1=1190$, $\sigma_1=90$, $n_1=1000$ | $x_2=1230$, $\sigma_2=120$, $n_2=75$

$$Z = (1190-1230) / \sqrt{(90^2/1000 + 120^2/75)} = -40 / \sqrt{(8.1+192)} = -40/14.15 = -2.83$$

$$|Z|=2.83 > 1.96 \rightarrow \text{REJECT } H_0 \rightarrow \text{Significant difference exists.}$$

TYPE 3 — Z-test: Difference of Two Proportions

HOW TO RECOGNISE: Two groups with proportions p_1 and p_2 . Test if $p_1 = p_2$.

$$\text{Pooled } P = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

$$Z = \frac{(p_1 - p_2) / \sqrt{[P \cdot Q \cdot (1/n_1 + 1/n_2)]}}{Q = 1 - P}$$

SOLVED EXAMPLE — D2 Apr 2025 — Referendum: 850 Men vs 560 Women

$$p_1 = 500/850 = 0.588, p_2 = 320/560 = 0.571$$

$$P = 820/1410 = 0.581, Q = 0.419$$

$$Z = (0.588 - 0.571) / \sqrt{[0.581 \times 0.419 \times (1/850 + 1/560)]} = 0.017 / 0.0269 = 0.63$$

$$|Z| = 0.63 < 2.576 \rightarrow \text{DO NOT REJECT } H_0 \rightarrow \text{No significant difference at 1\%}.$$

TYPE 4 — Paired t-test (Small Sample, Same Subjects)

HOW TO RECOGNISE: Two measurements on SAME subjects (e.g. before/after, husband/wife IQ). Compute $d = x_1 - x_2$.

$$d \text{ values} \quad d_i = x_{1i} - x_{2i} \text{ for each pair; } d = \sum d / n$$

$$S_d = \sqrt{[\sum (d - d)^2 / (n - 1)]} \text{ or } \sqrt{(\sum d^2 - n d^2) / (n - 1)}$$

$$t \text{ statistic} \quad t = d / (S_d / \sqrt{n}) \text{ df} = n - 1$$

SOLVED EXAMPLE — Apr 2023 G1 — IQ of Husbands vs Wives (n=10)

H: 117, 105, 97, 105, 123, 109, 86, 78, 103, 107

W: 106, 98, 87, 104, 116, 95, 90, 69, 108, 85

d: 11, 7, 10, 1, 7, 14, -4, 9, -5, 22

$$d = 72/10 = 7.2$$

$$\sum d^2 = 121 + 49 + 100 + 1 + 49 + 196 + 16 + 81 + 25 + 484 = 1122$$

$$S_d = \sqrt{[(1122 - 10 \times 51.84) / 9]} = \sqrt{[603.6 / 9]} = \sqrt{67.07} = 8.19$$

$$t = 7.2 / (8.19 / \sqrt{10}) = 7.2 / 2.59 = 2.78 \text{ df} = 9$$

$$\text{Critical } t_{0.05}(9) \text{ one-tail} = 1.833$$

$$t = 2.78 > 1.833 \rightarrow \text{REJECT } H_0 \rightarrow \text{Husbands have higher IQ.}$$

TYPE 5 — F-test for Equality of Variances

HOW TO RECOGNISE: Two samples. Test if $\sigma_1^2 = \sigma_2^2$. Put LARGER variance in numerator.

$$F = S_1^2 / S_2^2 \text{ (} S_1^2 \geq S_2^2 \text{) df: (} n_1 - 1, n_2 - 1 \text{)}$$

SOLVED EXAMPLE — D2 Apr 2025 — Two Normal Populations

A: $n_A=9$; B: $n_B=11$ — compute S_A^2 , S_B^2 from data, put larger on top.

$F = \max(S_A^2, S_B^2) / \min \rightarrow$ compare with $F_{\alpha, df_A, df_B}(8,10)$ from table.

If $F_{calc} < F_{table} \rightarrow$ Accept H_0 (variances are equal).

TYPE 6 — Chi-Square Test (Independence / Goodness of Fit)

HOW TO RECOGNISE: (a) Contingency table $\rightarrow$ test independence. (b) Observed freq vs theoretical dist $\rightarrow$ goodness of fit.

Test statistic $\chi^2 = \sum (O - E)^2 / E$

Expected (indep.) $E_{ij} = (\text{Row}_i \text{ total} \times \text{Col}_j \text{ total}) / \text{Grand total}$

df (independence) $(r-1)(c-1)$

df (goodness of fit) $k - 1 - (\text{number of estimated parameters})$

SOLVED EXAMPLE — Apr 2023 G1 — Hair Colour vs Eye Colour (3x3 table, N=150)

Hair: Fair(60), Brown(30), Black(60) Eye: Blue(40), Grey(50), Brown(60)

Expected values:

$E(\text{Blue, Fair}) = 40 \times 60 / 150 = 16$, $E(\text{Blue, Brown}) = 40 \times 30 / 150 = 8$, $E(\text{Blue, Black}) = 16$

$E(\text{Grey, Fair}) = 20$, $E(\text{Grey, Brown}) = 10$, $E(\text{Grey, Black}) = 20$

$E(\text{Brown, Fair}) = 24$, $E(\text{Brown, Brown}) = 12$, $E(\text{Brown, Black}) = 24$

$\chi^2 = (15-16)^2/16 + (5-8)^2/8 + (20-16)^2/16 + \dots$ (all 9 cells)

df = $(3-1)(3-1) = 4$; critical $\chi^2_{0.05, 4} = 9.49$

If $\chi^2_{calc} > 9.49 \rightarrow$ Hair and eye colour are associated.

TYPE 7 — One-Way ANOVA (Completely Randomised Design)

HOW TO RECOGNISE: k treatments, each with n observations. Test if all treatment means are equal.

CF $CF = G^2 / N$ G = grand total, N = total observations

SST $SST = \sum \sum x_{ij}^2 - CF$

SS(Tr) $SS(Tr) = \sum T_i^2 / n_i - CF$ T_i = treatment total

SSE $SSE = SST - SS(Tr)$

F $F = [SS(Tr)/(k-1)] / [SSE/(N-k)] = MS(Tr)/MSE$

SOLVED EXAMPLE — B2 Apr 2025 — CRD: 10 plots, 3 Treatments A/B/C

Treatments: A(plots 1,4,7,9), B(plots 2,8), C(plots 3,5,6,10)

Yields: A: 5,7,3,1=16; B: 4,4=8; C: 3,5,1,7=16

$N=10$, $G=40$, $CF=40^2/10=160$

$$\begin{aligned} SST &= (5^2+4^2+3^2+7^2+5^2+1^2+3^2+4^2+1^2+7^2)-160 \\ &= (25+16+9+49+25+1+9+16+1+49)-160 = 200-160 = 40 \end{aligned}$$

$$SS(Tr) = 16^2/4+8^2/2+16^2/4-160 = 64+32+64-160 = 0$$

$$SSE = 40-0 = 40$$

ANOVA Table:

Source | df | SS | MS | F

Treatments | 2 | 0 | 0 | 0

Error | 7 | 40 | 5.71 |

Total | 9 | 40 | |

$F=0 < F_{0.05}(2,7)=4.74 \rightarrow$ No significant treatment effect.

TYPE 8 — Two-Way ANOVA (Randomised Block Design)

HOW TO RECOGNISE: Rows = blocks/months, Columns = treatments/salesman. Test BOTH row and column effects.

$$SS(Rows) \quad SS(Rows) = \sum R_i^2/c - CF \quad R_i = \text{row total, } c = \text{no. of cols}$$

$$SS(Cols) \quad SS(Cols) = \sum C_j^2/r - CF \quad C_j = \text{col total, } r = \text{no. of rows}$$

$$SSE \quad SSE = SST - SS(Rows) - SS(Cols)$$

$$F(cols) \quad F = MS(Cols)/MSE \quad F(rows) = MS(Rows)/MSE$$

SOLVED EXAMPLE — D1 Apr 2025 — Refrigerator Sales: 3 Months × 4 Salesmen

Months (rows): March, April, May; Salesmen (cols): A, B, C, D $\rightarrow r=3$, $c=4$, $N=12$

Data: March: 50,40,48,39=177; April: 46,48,50,45=189; May: 39,44,40,39=162

Col totals: A=135, B=132, C=138, D=123

$G = 528$, $CF = 528^2/12 = 23232$

$$SST = (50^2+40^2+...+39^2)-23232$$

$$SS(Months) = (177^2+189^2+162^2)/4 - 23232$$

$$SS(Salesman) = (135^2+132^2+138^2+123^2)/3 - 23232$$

$$SSE = SST - SS(Months) - SS(Salesman)$$

Test both F values at 5% level.

TOPIC 7

Reliability Engineering

Hazard Function · Failure PDF · Series/Parallel · Availability

NOTE: The $\lambda(t)=0.003(t/500)^{0.5}$ problem appeared in B2/B1/D2 Apr 2025 and D2 May 2024. The availability problem ($\lambda=0.1$, $MTTR=2.5$) appeared in G1/B1 Apr 2025. These are your top-priority reliability problems.

TYPE 1 — Given Hazard Function $\lambda(t)$: Find $R(t)$, MTTF, Design Life ★

HOW TO RECOGNISE: $\lambda(t)$ = some function of t given. Find reliability, MTTF, and design life for a target reliability.

Key Formulas

Reliability	$R(t) = \exp\left[-\int_0^t \lambda(u) \, du\right]$
MTTF	$MTTF = \int_0^\infty R(t) \, dt$
Design life	Solve $R(t_d) = R_{\text{required}}$ for t_d
Conditional R	$P(T \geq t_1 + t_2 \mid T \geq t_1) = R(t_1 + t_2) / R(t_1)$
Annual maint.	$R_{\text{effective}}(nT) = [R(T)]^n$ (system reset each period T)

SOLVED EXAMPLE — ★ $\lambda(t) = 0.003(t/500)^{0.5}$ [APPEARED IN 4 PAPERS]

Compute $\int_0^t \lambda(u) du$:

$$= \int_0^t 0.003(u/500)^{0.5} du = 0.003/\sqrt{500} \times \int_0^t u^{0.5} du$$

$$= (0.003/22.36) \times (2/3)t^{3/2} = 0.0000894 \times t^{3/2}$$

$$\Rightarrow R(t) = \exp[-0.0000894 \cdot t^{3/2}]$$

(i) $R(50)$:

$$= \exp[-0.0000894 \times 50^{3/2}] = \exp[-0.0000894 \times 353.6] = \exp[-0.03160] = 0.9689$$

(ii) Design life for $R=0.90$:

$$\exp[-0.0000894 \cdot t^{3/2}] = 0.90$$

$$-0.0000894 \cdot t^{3/2} = \ln(0.90) = -0.10536$$

$$t^{3/2} = 1178.6 \rightarrow t = (1178.6)^{2/3} \approx 115.5 \text{ hours}$$

(iii) MTTF: Recognise as Weibull with $\beta=1.5$

Numerically integrate $R(t)$ from 0 to ∞ , or use Γ function.

(iv) $P(T \geq 100 | T \geq 50) = R(100)/R(50)$:

$$R(100) = \exp[-0.0000894 \times 1000] = \exp[-0.0894] = 0.9145$$

$$\Rightarrow \text{Conditional } R = 0.9145/0.9689 = 0.944$$

SOLVED EXAMPLE — D1 Apr 2025 — $\lambda(t) = 0.23 + 0.01t^2$ (Wear-out machine)

$$\int_0^t \lambda(u) du = 0.23t + 0.01t^3/3$$

(i) $R(3)$ without maintenance:

$$= \exp[-(0.23 \times 3 + 0.01 \times 27/3)] = \exp[-(0.69 + 0.09)] = \exp[-0.78] = 0.458$$

(ii) $R(3)$ with annual maintenance (reset every year):

$$R(1) = \exp[-(0.23 + 0.01/3)] = \exp[-0.2333] = 0.792$$

$$R_{\text{effective}}(3) = [R(1)]^3 = (0.792)^3 = 0.497$$

Annual maintenance IMPROVES reliability: $0.497 > 0.458$ ✓

TYPE 2 — Given Failure PDF $f(t)$: Find $R(t)$, $\lambda(t)$, MTTF

HOW TO RECOGNISE: $f(t) = 200/(t+10)^3$ or $t^2/333$ etc. Derive $R(t)$ by integration, then $\lambda(t) = f(t)/R(t)$.

Key Formulas

CDF	$F(t) = \int_0^t f(u) du$
Reliability	$R(t) = 1 - F(t) = \int_t^\infty f(u) du$
Hazard rate	$\lambda(t) = f(t) / R(t)$

MTTF

$$MTTF = \int_0^{\infty} R(t) dt$$

Burn-in

$$R_{\text{new}}(t) = R(t+t_0) / R(t_0) \text{ [after burn-in of } t_0 \text{]}$$

SOLVED EXAMPLE — G1 2024 / B1 2024 — $f(t) = 200/(t+10)^3$, $t \geq 0$

$$F(t) = \int_0^t 200/(u+10)^3 du = [-100/(u+10)^2]_0^t = 1 - 100/(t+10)^2$$

$$R(t) = 100/(t+10)^2$$

(i) $R(1 \text{ year})$:

$$R(1) = 100/121 = 0.826$$

$$(ii) MTTF = \int_0^{\infty} 100/(t+10)^2 dt = 100 \times [-1/(t+10)]_0^{\infty} = 100/10 = 10 \text{ years}$$

(iii) Design life for $R=0.95$:

$$100/(t+10)^2 = 0.95 \rightarrow (t+10)^2 = 105.26 \rightarrow t = 0.26 \text{ years}$$

$$(iv) \lambda(t) = f(t)/R(t) = [200/(t+10)^3] / [100/(t+10)^2] = 2/(t+10)$$

λ is increasing $\rightarrow$ WEAR-OUT failure mode ✓

(v) One-year burn-in: $R_{\text{new}}(1) = R(1+1)/R(1) = R(2)/R(0.826)$

$$R(2) = 100/144 = 0.694; R_{\text{new}}(1) = (100/144)/(100/121) = 121/144 = 0.840$$

Burn-in IMPROVES reliability ($0.840 > 0.826$) ✓

SOLVED EXAMPLE — Apr 2023 B1 — Repair PDF $m(t) = t^2/456$, $1 \leq t \leq 20$ [MAINTAINABILITY]

Note: This is a REPAIR time PDF, not failure! Answers give MTTR and repair rate $\mu(t)$.

$$MTTR = \int_1^{20} t \cdot (t^2/456) dt = (1/456) \int_1^{20} t^3 dt = (1/456) [t^4/4]_1^{20}$$

$$= (1/1824)(160000-1) = 159999/1824 \approx 87.7 \text{ time units}$$

$$M(t) = \int_1^t u^2/456 du = [u^3/1368]_1^t = (t^3-1)/1368$$

$$\text{Repair rate } \mu(t) = m(t)/(1-M(t)) = (t^2/456) / [1-(t^3-1)/1368]$$

$$P(\text{repair by } 12) = M(12) = (1728-1)/1368 = 1727/1368 = 1.26 > 1?$$

Recheck normalisation — the exact constant may differ in the paper.

TYPE 3 — System Reliability: Series and Parallel

HOW TO RECOGNISE: Component reliabilities $R_1, R_2, \dots$ given. Find system reliability for series/parallel configuration.

Key Formulas

Series

$$R_{\text{sys}} = R_1 \times R_2 \times \dots \times R_n \text{ [all must work]}$$

Parallel

$$R_{\text{sys}} = 1 - (1-R_1)(1-R_2)\dots(1-R_n) \text{ [at least one must work]}$$

Exponential
comp.

$$R_i(t) = e^{(-\lambda_i t)}$$

Series + exp.

$$R_{sys}(t) = e^{(-\lambda_{sys} \times t)} \quad \lambda_{sys} = \sum \lambda_i \times (\text{count of each})$$

SOLVED EXAMPLE — D2 Apr 2025 — Series Circuit: 5 Transistors + 3 Diodes + 10 Resistors + 2 Caps, t=10h

Failure rates: $\lambda_t = 4 \times 10^{-6}$, $\lambda_d = 3 \times 10^{-6}$, $\lambda_r = 2 \times 10^{-6}$, $\lambda_c = 2 \times 10^{-6}$

Total system failure rate (series):

$$\begin{aligned} \lambda_{sys} &= 5(4 \times 10^{-6}) + 3(3 \times 10^{-6}) + 10(2 \times 10^{-6}) + 2(2 \times 10^{-6}) \\ &= 20 \times 10^{-6} + 9 \times 10^{-6} + 20 \times 10^{-6} + 4 \times 10^{-6} = 269 \times 10^{-6} \end{aligned}$$

$$R_{sys}(10) = \exp(-269 \times 10^{-6} \times 10) = \exp(-0.0269) = 0.9734$$

SOLVED EXAMPLE — Apr 2023 G1 — 3 Parallel Components R=0.20, 0.40, 0.50

$$\text{Baseline: } R_{sys} = 1 - (0.80)(0.60)(0.50) = 1 - 0.24 = 0.76$$

Case (i) Increase R to 0.30:

$$R = 1 - (0.70)(0.60)(0.50) = 1 - 0.21 = 0.79 \rightarrow \% \text{ increase} = 3/0.76 \times 100 = 3.95\%$$

Case (ii) Increase R to 0.50:

$$R = 1 - (0.80)(0.50)(0.50) = 1 - 0.20 = 0.80 \rightarrow \% \text{ increase} = 5.26\%$$

Case (iii) Increase R to 0.60:

$$R = 1 - (0.80)(0.60)(0.40) = 1 - 0.192 = 0.808 \rightarrow \% \text{ increase} = 6.3\%$$

=> Increasing R gives the largest improvement.

TYPE 4 — System Availability (Point, Interval, Steady-State) ★

HOW TO RECOGNISE: Failure rate λ and mean repair time MTTR given. Compute various availability measures.

Key Formulas

Repair rate

$$\mu = 1/\text{MTTR}$$

Point avail.

$$A(t) = \mu/(\lambda + \mu) + [\lambda/(\lambda + \mu)] \cdot e^{-(\lambda + \mu)t}$$

Steady-state

$$A_{ss} = \mu/(\lambda + \mu)$$

Interval avail.

$$\bar{A} = (1/T) \cdot \int_0^T A(t) dt$$

Series (2)

$$A_{sys}(t) = [A(t)]^2$$

Parallel (2)

$$A_{ss} = 1 - (1 - A_{ss})^2$$

Standby (no
fail)

$$A_{ss} = (\lambda + 2\mu) / (2(\lambda + \mu))$$

SOLVED EXAMPLE — ★ G1/B1 Apr 2025, B1 May 2024 — $\lambda=0.1/\text{day}$, MTTR=2.5 days [SAME PROBLEM!]

Setup: $\mu = 1/2.5 = 0.4$, $\lambda + \mu = 0.5$

(i) Point availability at $t=2$:

$$A(2) = 0.4/0.5 + (0.1/0.5) \cdot e^{-(0.5 \times 2)} = 0.8 + 0.2e^{-1} = 0.8 + 0.2 \times 0.368 = 0.874$$

Interval availability over 2-day mission:

$$\begin{aligned} \blacksquare &= (1/2) \cdot \int_0^2 A(t) dt = (1/2) \cdot [0.8t - (0.2/0.5)e^{-(0.5t)}] \blacksquare^2 \\ &= (1/2) \cdot [1.6 - 0.4(e^{-1} - 1)] = (1/2) \cdot [1.6 + 0.4 - 0.147] = 0.927 \end{aligned}$$

Steady-state: $A_{ss} = 0.4/0.5 = 0.80$

(ii) Two relays in SERIES at $t=2$:

$$A_{sys}(2) = [A(2)]^2 = (0.874)^2 = 0.764$$

(iii) Two relays in PARALLEL (steady-state):

$$A_{ss} = 1 - (1 - 0.80)^2 = 1 - 0.04 = 0.96$$

(iv) Standby mode (no failure in standby):

$$A_{ss} = (\lambda + 2\mu) / (2\lambda + 2\mu) = (0.1 + 0.8) / (0.2 + 0.8) = 0.9/1.0 = 0.90$$

VARIATION SUMMARY TABLE

Quick-reference for the exam. For each type: what triggers it, and the single most important trick.

Topic	Type	Recognition Trigger	Key Trick / Formula
Descriptive Stats	1: Mean/Median/Mode (grouped)	Class interval + freq table	$L + (N/2 - CF)/f \times h$ for Median; mode uses $f_1 - f_0$ in numerator
	2: SD+QD+CV+Skewness	Wage/salary table, ALL measures asked	$\sigma = \sqrt{(\sum f m^2 / N - x^2)}$; $QD = (Q_3 - Q_1)/2$; $CV = \sigma / x \times 100$
	3: Missing frequency	f_i missing, Mean or Median given	Form equation from mean formula; solve simultaneously
	4: CV comparison	Which team/group is MORE CONSISTENT?	Compute CV for both; LOWER CV = more consistent
Discrete RV	1: Find k, Mean, Var, CDF	PMF table with k in it	$\sum p(x) = 1 \rightarrow k$; $E = \sum xp$; $Var = E(X^2) - [E(X)]^2$
	2: Geometric / Infinite dist.	$P(X=x)$ = formula; $x=1,2,3,\dots$	MEMORYLESS: $P(X>a+b X>a) = P(X>b)$
	3: $Y = g(X)$ function	$Y = X^2 + 2X$ or similar transform	Compute Y for each x; build new PMF; $E(Y) = \sum y \cdot P(Y)$
	4: Continuous PDF	$f(x) = k \cdot x^n$ over $[0,1]$	$\int f dx = 1 \rightarrow k$; $P = \int f dx$; $CDF = \int f dt$
Joint Dist.	1: Discrete joint table	2D table of $P(X=x, Y=y)$	Row sums=marginal X; col sums=marginal Y; $Cov = E(XY) - E(X)E(Y)$
	2: ★ Continuous (rectangle)	$f(x,y) = x^2 + xy/3$ over rectangle	Marginal = integrate out other variable; $P(Y<X)$ uses y:0 to x limit
	3: Continuous (triangle)	$0 < x < y < 1$ type region	CRITICAL: $f_X(x) \rightarrow y$ limit is x to 1, NOT 0 to 1. Sketch first!
Correlation & Regression	1: Pearson's r	Raw (x,y) data table	$r = [n\sum xy - \sum x \sum y] / \sqrt{[(n\sum x^2 - (\sum x)^2)(n\sum y^2 - (\sum y)^2)]}$
	2: Rank correlation	Data with possible ties	Ties: use avg rank; add $CF = m(m^2 - 1)/12$ correction
	3: Two regression lines	Estimate Y for $X = \text{val}$ AND X for $Y = \text{val}$	Both lines pass through $(\bar{x}, \bar{y})$; check $b_{yx} \times b_{xy} = r^2$
	4: ★ Destroyed lab record	$Var(X) = 9$, two regression equations	Solve for means; identify b_{yx} vs b_{xy} ; $r = \sqrt{(\text{product})}$; σ_Y from b_{yx} formula
Distributions	1: Binomial	n fixed trials, p fixed, count successes	$P(X=r) = C(n,r) p^r q^{n-r}$; $E = np$; $Var = npq$
	2: Poisson	λ given or estimated from data (rare events)	$P(X=k) = e^{-\lambda} \lambda^k / k!$; for GOF: estimate λ , compute expected, run χ^2
	3: Normal	μ, σ given; convert to $Z = (X - \mu) / \sigma$	Find μ, σ from TWO conditions: set up two equations from Z-table values
Hypothesis Testing	1: Z-test single mean	Large n, test $\mu = \mu_0$	$Z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$; $ Z > 1.96$ at 5% two-tail
	2: Z-test two means	Two large samples, compare means	$Z = (\bar{x}_1 - \bar{x}_2 - (\mu_1 - \mu_2)) / \sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$

Topic	Type	Recognition Trigger	Key Trick / Formula
	3: Z-test proportions	Two groups with proportions p_1, p_2	Pool: $P = (n_1 p_1 + n_2 p_2) / (n_1 + n_2)$; $Z = (p_1 - p_2) / \sqrt{P Q (1/n_1 + 1/n_2)}$
	4: Paired t-test	SAME subjects measured twice	$d = x_1 - x_2$; $t = d / (S_d / \sqrt{n})$; $df = n - 1$
	5: F-test	Compare two sample variances	$F = \text{larger } S^2 / \text{smaller } S^2$; always put larger on top
	6: Chi-square	Contingency table / fit distribution	$E = \text{RowTotal} \times \text{ColTotal} / N$; $df = (r-1)(c-1)$
	7: One-way ANOVA	k treatments, test equality of means	$F = MS(\text{Tr}) / MSE$ where $SS(\text{Tr}) = \sum T_i^2 / n_i - CF$
	8: Two-way ANOVA	Rows=blocks, cols=treatments	Add $SS(\text{Rows})$ and $SS(\text{Cols})$ separately; test both F values
Reliability	1: ★ Given $\lambda(t)$	Hazard rate function given	$R(t) = \exp(-\int \lambda dt)$; design life: solve $R(t_d) = R_{\text{target}}$ for t_d
	2: Given $f(t)$	PDF of failure time given	$R(t) = \int_t^\infty f du$; $\lambda(t) = f(t) / R(t)$; check if increasing (wear-out)
	3: Series/Parallel	Component R or λ values given	Series: $\prod R_i$; Parallel: $1 - \prod (1 - R_i)$; Exp series: sum λ 's
	4: ★ Availability	λ and MTTR given	$\mu = 1 / \text{MTTR}$; $A_{ss} = \mu / (\lambda + \mu)$; point $A(t)$ has $e^{-(\lambda + \mu)t}$ term

FINAL EXAM TIPS

★ Must-memorise (4+ appearances each):	$f(x,y) = x^2 + xy/3$ joint PDF Destroyed lab record (Var $X=9$) $\lambda(t) = 0.003(t/500)^{0.5}$ Availability with $\lambda=0.1$, MTTR=2.5
Joint PDF region:	ALWAYS sketch region before integrating. $0 < x < y < 1$ is a triangle — limits change!
Regression lines:	ALWAYS find BOTH lines unless the question says otherwise. Use intersection point to find means.
t-test vs Z-test:	$n \leq 30$ AND σ unknown $\rightarrow$ t-test. $n > 30$ OR σ known $\rightarrow$ Z-test.
Paired vs 2-sample:	SAME people measured twice $\rightarrow$ Paired t-test. DIFFERENT people $\rightarrow$ Two-sample t-test.
ANOVA vs t-test:	2 groups $\rightarrow$ t-test. 3+ groups $\rightarrow$ ANOVA. One factor $\rightarrow$ one-way. Two factors $\rightarrow$ two-way.
Rank correlation ties:	Assign AVERAGE rank to tied observations, then add $CF = m(m^2-1)/12$ for each tie group.
Normal distribution:	If asked to find μ and σ — set up TWO equations from two probability conditions and solve.

— *End of Study Guide* —